Artificial Intelligence for Digital Characters

Speech Synthesis I

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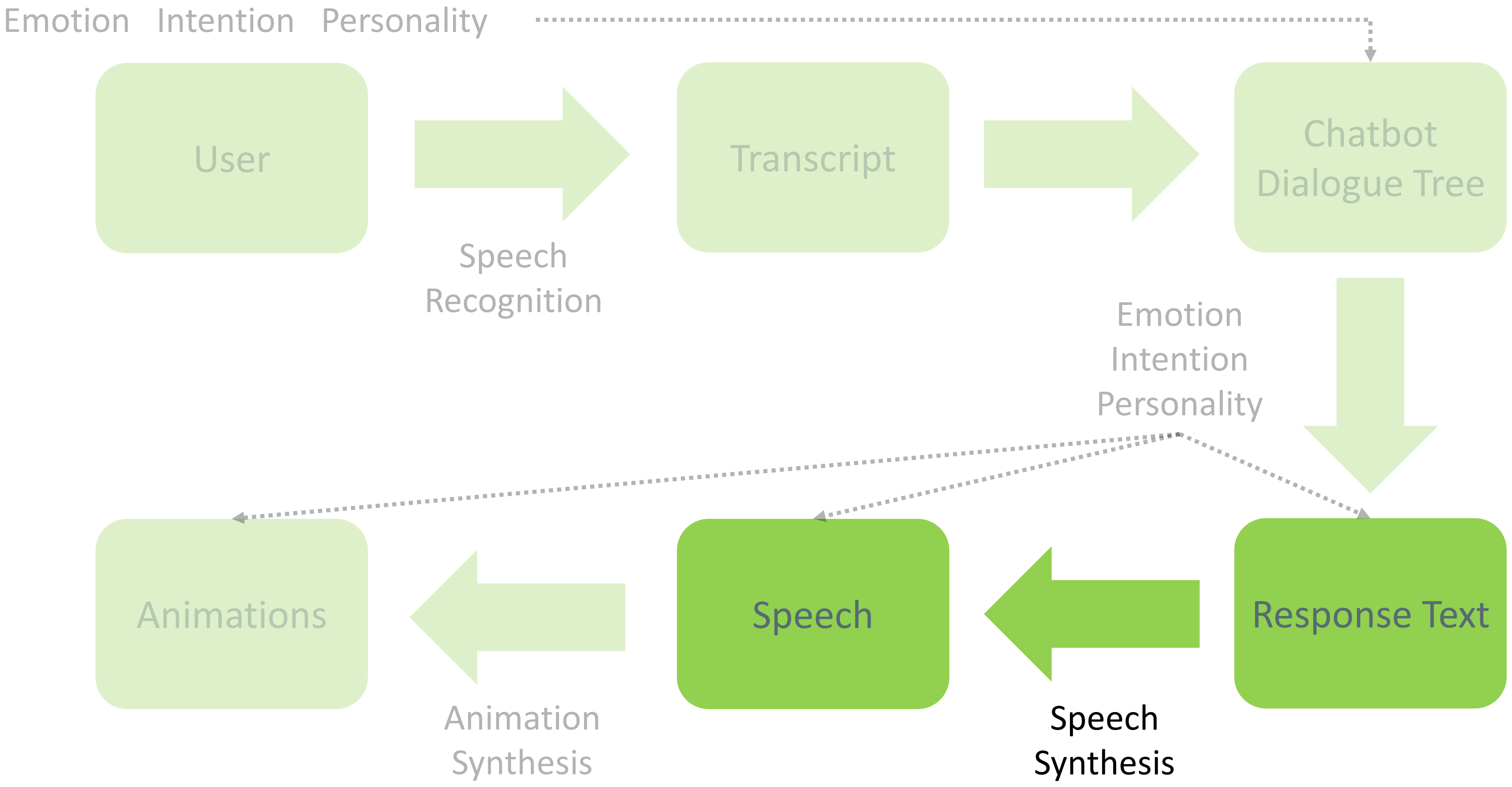
Conversational Digital Character



Speech Synthesis Is Used Everywhere

- Conversational Agents
- Devices reading out loud for the blind
- Games
- People suffering from medical issues

History

Text Normalization
Phonetic Analysis
Prosodic Analysis
Diphone Synthesis
Unit Selection Synthesis

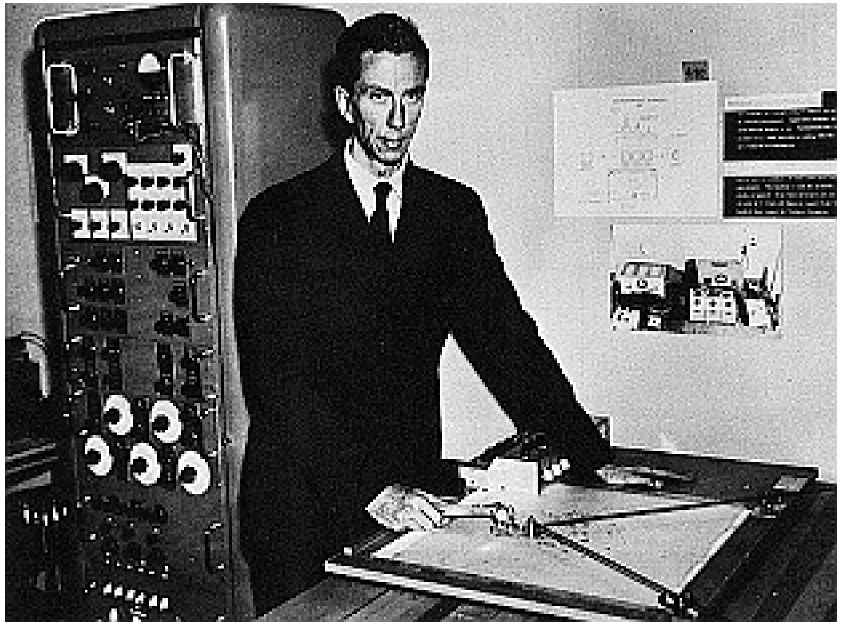
Early Systems
Types of Synthesis

Homer Dudley VODER

- Synthesizing speech by electrical means
- 1939 World’s Fair

Gunnar Fant's OVE synthesizer

Of the Royal Institute of Technology, Stockholm



Pre-modern TTS systems

- 1960’s first full TTS: Umeda et al (1968)
- 1970’s
  - Joe Olive 1977 concatenation of diphones
  - Texas Instruments Speak and Spell (1978)

- Were the most common commercial systems when computers were slow and had little memory.
- 1979 MIT MITalk (Allen, Hunnicut, Klatt)
- 1983 DECtalk system
  - "Perfect Paul" (The voice of Stephen Hawking)
  - "Beautiful Betty"

Parametric Synthesis

- Train a statistical model on large amounts of data
  Learn text → sound mapping
- Deep learning approaches are the latest generation of parametric
  - Quickly becoming adopted in industry due to improved naturalness and compact models

[Google audiobook TTS]

History

Text Normalization

Phonetic Analysis

Prosodic Analysis

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Unit Selection Synthesis

Tokens

Chunks

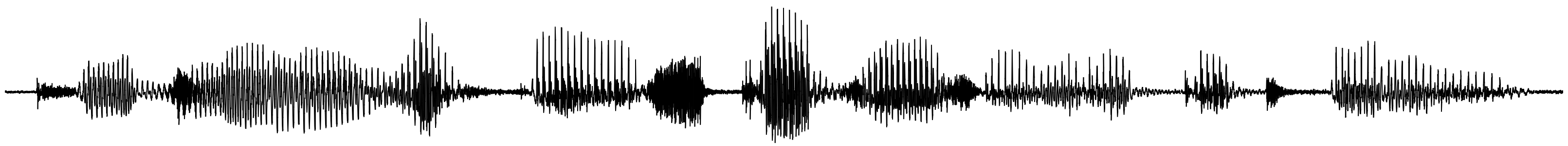
Homographs

PG&E will file schedules on April 20.

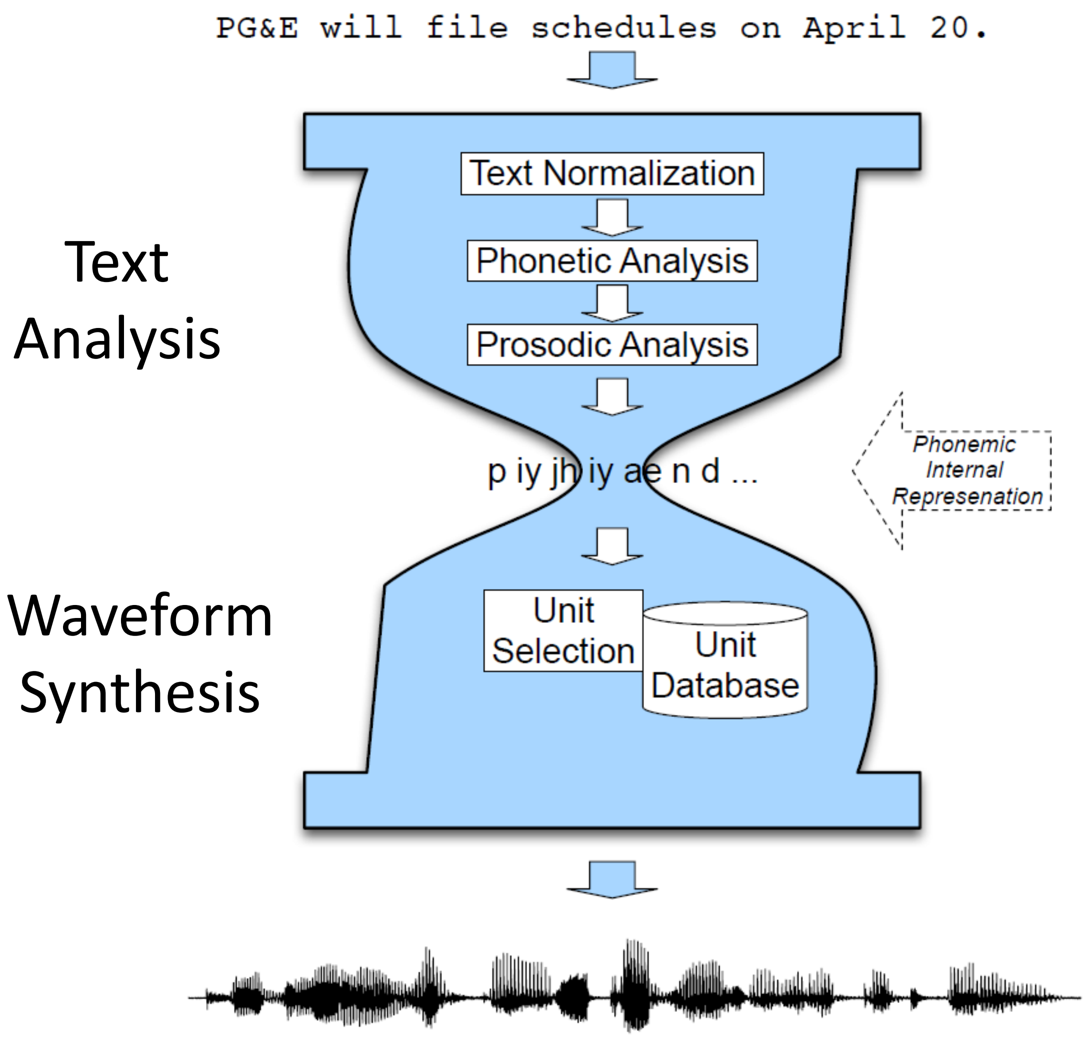
1. **Text Analysis**: Text into intermediate representation:

			*			*			* L-L%
P	G	AND	E	WILL	FILE	SCHEDULES	ON	APRIL	TWENTIETH
p iy	jh iy	ae n d	iy	w ih l	f ay l	s k eh jh ax l z	aa n	ey p r ih l	t w eh n t iy ax th

2. **Waveform Synthesis**: From the intermediate representation into waveform



The two stages of TTS



Text Normalization

Analysis of raw text into pronounceable words:

He said the increase in credit limits helped B.C. Hydro achieve record net income of about $1 billion during the year ending March 31. This figure does not include any write-downs that may occur if Powerex determines that any of its customer accounts are not collectible. Cousins, however, was insistent that all debts will be collected: "We continue to pursue monies owing and we expect to be paid for electricity we have sold."

Text Normalization

- He stole $100 million from the bank
- It’s 13 St. Andrews St.
- Yes, see you the following tues, that’s 11/12/01
- IV: four, fourth, I.V.
- 1750: seventeen fifty (date, address) or one thousand seven… (dollars)

Text Normalization Steps

1. Identify tokens in text
2. Chunk tokens into reasonably sized sections (sentences)
3. Identify types of tokens
4. Convert tokens to words

Step 1+2: Identify Tokens and Chunk

- Identification of tokens
  - Whitespace can be viewed as separators
  - Punctuation can be separated from the raw tokens
- End of sentence detection
  - Relatively simple if utterance ends in ?!
  - Ambiguity for period
    My place on Main St. is around the corner.
    I live at 123 Main St. (Not I live at 123 Main St..)
  - Use machine learning to predict if token is end of sentence

Steps 3+4: Identify Types of Tokens, and Convert Tokens to Words

- Pronunciation of numbers often depends on type. Three ways to pronounce 1776:
  - **Date**: seventeen seventy six
  - **Phone number**: one seven seven six
  - **Quantifier**: one thousand seven hundred (and) seventy six
- Non-standard word types
  - **Abbreviations**: expansion (Jan 1 = January first)
  - **Letter sequence**: pronouncing each letter (IBM = ay b iy eh m)
  - **Acronyms**: Pronounced as if they were words (e.g., IKEA)

Steps 3+4: Identify Types of Tokens, and Convert Tokens to Words

- Identifying type of token
  - Any word not in pronunciation dictionary is non-standard word
  - Regular expressions
    e.g., */(1[89][0-9][0-9])|(20[0-9][0-9]/* for detecting years
  - Machine learning classifier
- Convert tokens to words
  - Expanding by using an abbreviation dictionary

Homograph Disambiguation

Do you live (/l ih v/) near a zoo with live (/l ay v/) animals?

1. Is token in homograph dictionary?
2. Recognize part-of-speech (verb, noun, adjective,…)
3. Look up of pronunciation in dictionary

History

Text Normalization

Phonetic Analysis

Prosodic Analysis

Diphone Synthesis

Unit Selection Synthesis

- Dictionaries
- Names
- Grapheme-to-Phoneme

Produce pronounciation for each word (LTS)

- Big dictionary
- Special code for handling names
- Machine learning LTS system for other unknown words

Dictionaries

- CMU dictionary
  - 127K words
  - 39-phone phoneme set
  - Stress tag (vowels): 0 (unstressed), 1 (stressed), or 2 (secondary stress)
  - http://www.speech.cs.cmu.edu/cgi-bin/cmudict

ANTECEDENTS	AE2 N T IH0 S IY1 D AH0 N T S	*PAKISTANI*	P AE2 K IH0 S T AE1 N IY0
CHANG	CH AE1 NG	*TABLE*	T EY1 B AH0 L
DICTIONARY	D IH1 K SH AH0 N EH2 R IY0	*TROTSKY*	T R AA1 T S K IY2
DINNER	D IH1 N ER0	*WALTER*	W AO1 L T ER0
LUNCH	L AH1 N CH	*WALTZING*	W AO1 L T S IH0 NG
MCFARLAND	M AH0 K F AA1 R L AH0 N D	*WALTZING(2)*	W AO1 L S IH0 NG

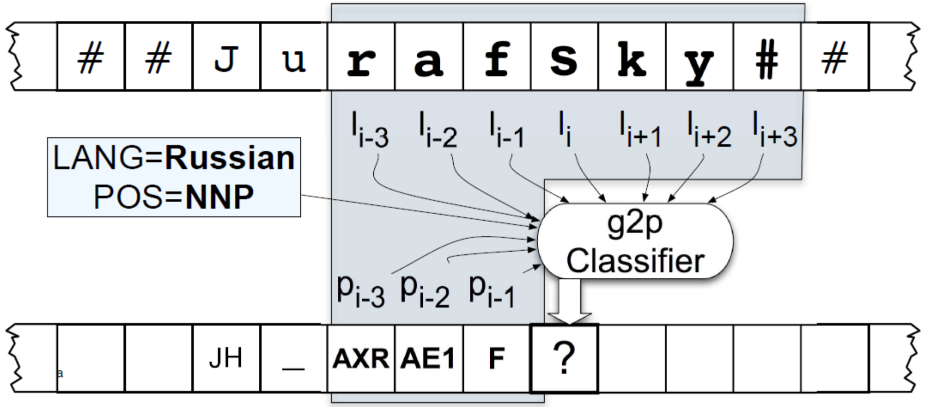
- Unisyn dictionary
  - 110K words
  - Significantly more accurate, includes multiple dialects
  - http://www.cstr.ed.ac.uk/projects/unisyn/

Names

- 20% of tokens in typical newswire text
  - Personal names (McArthur, Jiminez, etc.)
  - Company/Brand names (IKEA, IBM, etc.)
- Can do morphology
  - Walter = Walter + s
  - Lucasville = Lucas + ville
- Rhyme analogy: **Pl**otsky by analogy with Trotsky
- Can do automatic country detection and then do country-specific rules

Grapheme-to-Phoneme

- Converting a sequence of letters into a sequence of phones (letter-to-sound)
- Earlier work: rule-based
- Machine learning classifier
  - Window of k letters
  - k produced phones
  - Separate classifier for unknown names and other unknown words



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Prominence & Accent

Boundaries

Duration

F0

Prosody and Intonation in TTS

- **Prominence/Accent**: Decide which words are accented, which syllable has accent, what sort of accent
- **Boundaries**: Decide where intonational boundaries are
- **Duration**: Specify length of each segment
- **F0**: Generate F0 contour from these

Stress vs. Accent

- **Accent:** property of a word in context
  - Context-dependent. Marks important words in the discourse
- **Stress**: structural property of a word
  - Fixed in the lexicon: marks a potential location for an accent to occur, if there is one

(x)					(x)		(accented word)
x					x		stressed syllable
x	x	x	x	x	x	x	syllables
vi	ta	mins	Ca	li	for	nia	

Levels of Prominence

- Most phrases have more than one accent
- **Nuclear Accent:** Last accent in a phrase, perceived as more prominent
  - Plays semantic role like indicating a word is contrastive or focus.
  - "I know **SOMETHING** interesting is sure to happen’ she said
- Can also have reduced words that are **less** prominent than usual (especially function words)
- Typically use 4 classes of prominence:
  - Unaccented, reduced, pitch accent, empathic (nuclear) accent

Levels of Prominence

- Avoid putting accents to close together (clash) or too far apart (lapse)
  - City **HALL PARKING** lot → **CITY** hall **PARKING** lot
- Machine learning to predict accents

Boundaries

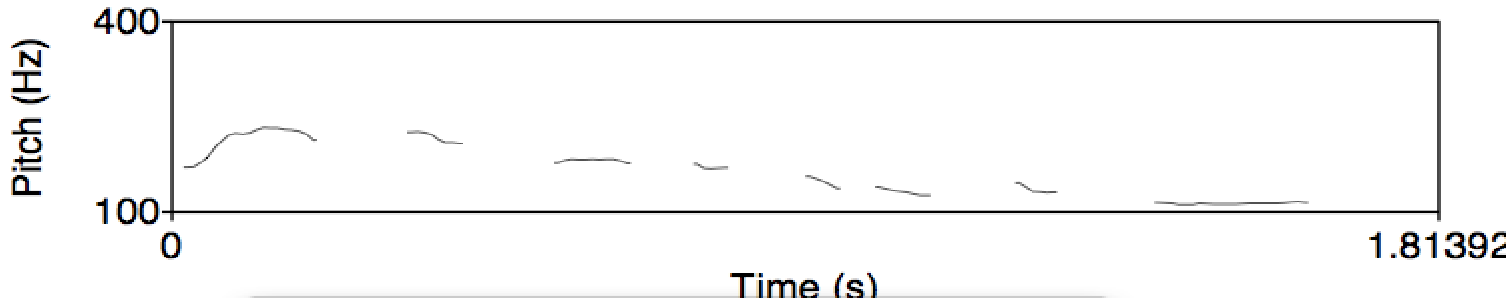
- Full || (intonation phrase) versus intermediate |
  - Doctor Norman Rosenblatt, || dean of the college | of criminal justice at Northeastern University, || agrees.
- Implications
  - Final vowel of a intonation phrase longer than usual
  - Pause after an intonation phrase
  - Slight drop in F0 from the beginning of an intonation phrase to its end
- Classification
  - Rules: boundary after punctuation, before function word following content word
  - Machine learning

Duration & F0

- Phone duration
  - Rule-based
  - Machine learning
- F0
  - Predicting F0 target points for each syllable

F0 Generation by Rule

- Generate a list of target F0 points for each syllable
- Example
  - Accented word *hat*: 3 pitch points [110, 140, 100]
- Modified by gender, declination, end of sentence



F0 Generation by Machine Learning

- Supervised machine learning
- Predict: value of F0 at 3 places in each syllable (start, middle, end)
- Predictor features:
  - Accent of current word, next word, previous word
  - Boundaries
  - Stress information
- Need training sets with labels
- F0 is generally defined relative to pitch range
  - Range between baseline and topline frequency in an utterance

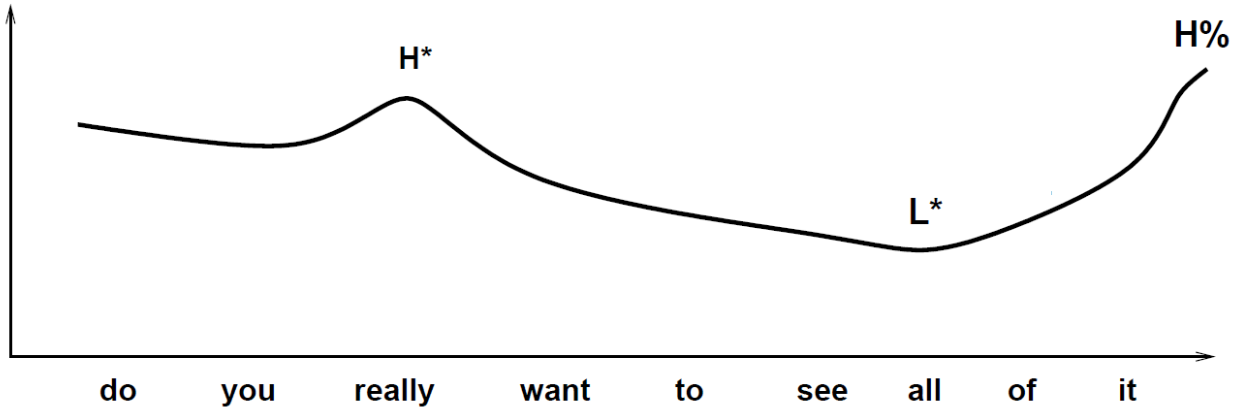
Final Result

H*: Peak accent

L*: Low accent

L-H%: Continuation rise

				H*												L*		L- H%			
do		you		really				want				to		see		all		of		it	
d	uw	y	uw	r	ih	l	iy	w	aa	n	t	t	ax	s	iy	ao	l	ah	v	ih	t
110	110	50	50	75	64	57	82	57	50	72	41	43	47	54	130	76	90	44	62	46	220



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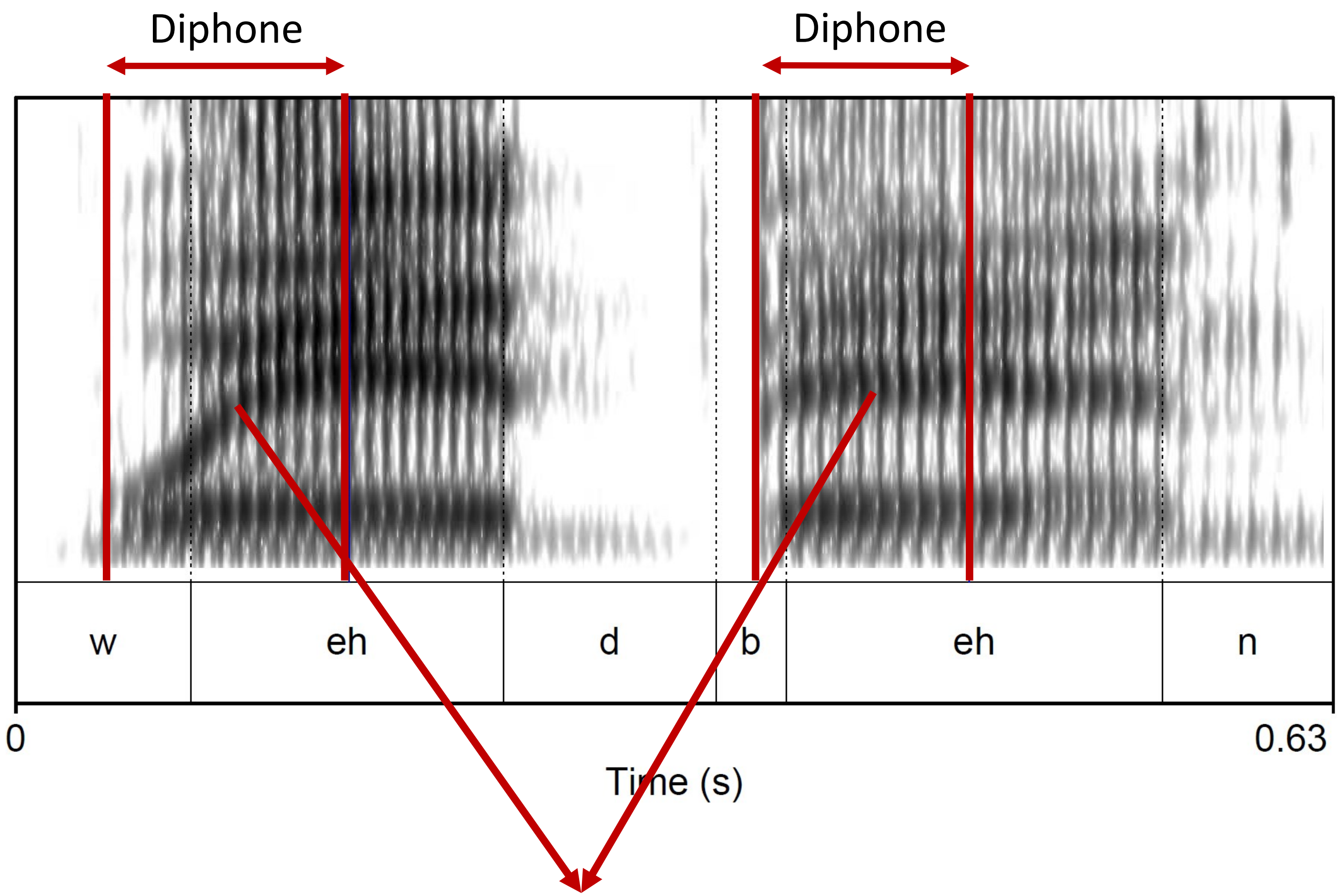
Diphone Synthesis
Unit Selection Synthesis

Diphones
Architecture
Epochs

Diphones



Phone [eh]: Second formant (F2) depends on previous and next phone

- Coarticulation: Each phone differs depending on previous and following phone
- Mid-phone is more stable than edge (less influenced by context)
- Diphone: Middle of phone to middle of next phone

Diphones

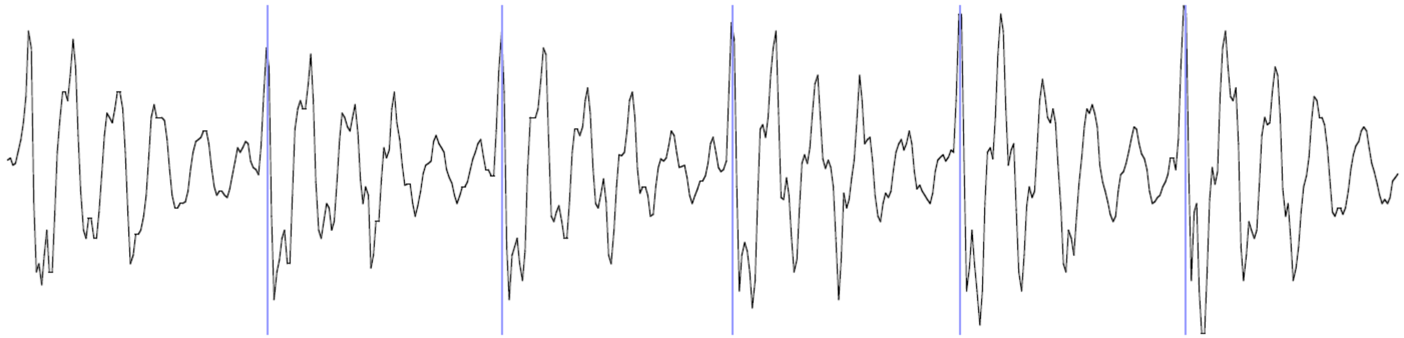
- Let's assume 39 phones
- $39^2 = 1521$ diphones
  - Some combinations do not exist
  - [h], [y], [w] only occurs before vowels
  - Do not need to keep diphones if no coarticulation between phones (e.g., silence)
  - This results in 1172 actual diphones
- Database relatively small
  - Around 8 megabytes for English (16 KHz, 16 bit)

Diphone TTS Architecture

- Training
  - Record a speaker saying one example sentence of each diphone
  - Segment the phones (speech recognizer & manual correction)
  - Mark the boundaries of each diphone and cut diphone out (store in database)
    - Rule-based: 50% into the phone, for stops 30%
    - Optimal coupling: Decide cutting point at runtime so that final frame of first diphone is acoustically most similar to the end frame of the next diphone
- Synthesizing an utterance
  - Retrieve sequence of diphones from database
  - Concatenate the diphones
    - Apply windowing function (Hanning window) at edges of diphones (low amplitude at juncture)
    - Pitch at end of first diphone should be similar to pitch at beginning of second diphone
  - Use signal processing to change the prosody (F0, energy, duration) of diphone sequence

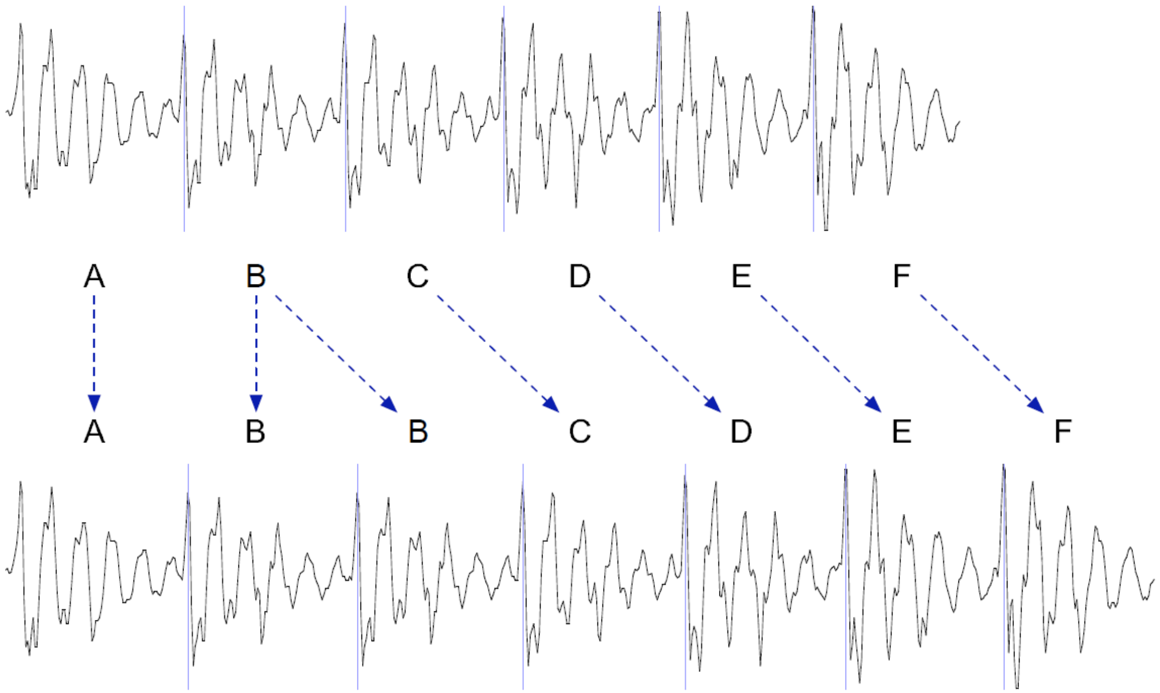
Pitch marking (epoch labeling)

- Specific points within each cycle of vibration of the vocal folds
- Different algorithms for epoch detection exist
- Visualization with *Show Pulses* in Praat

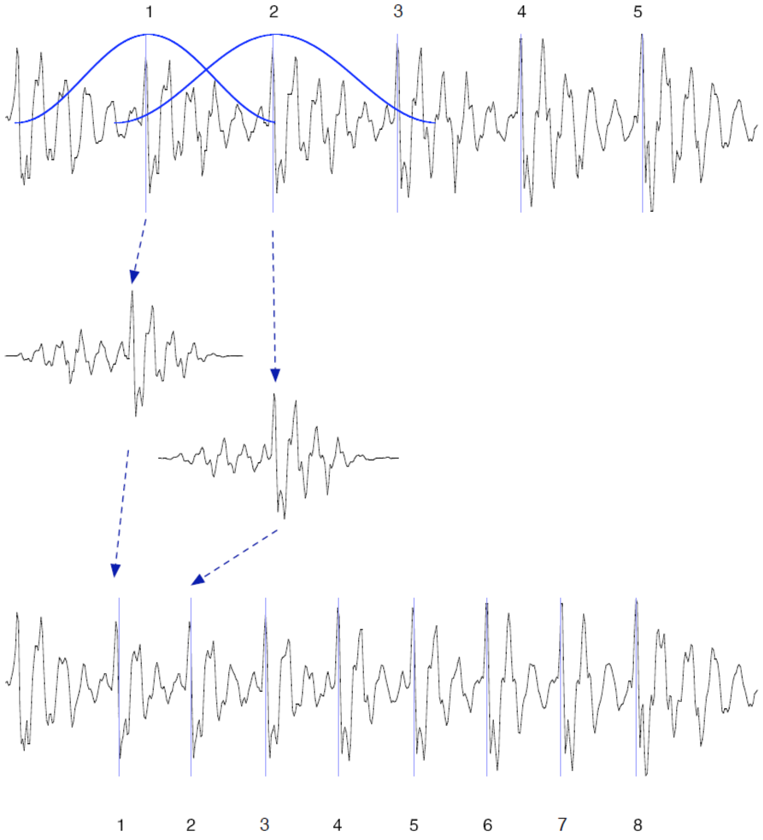


Duration & Pitch Modifications

Increasing duration
(duplication of frames)



Increasing F0
(Overlapping frames, adding up, duplication)



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Unit Selection Synthesis

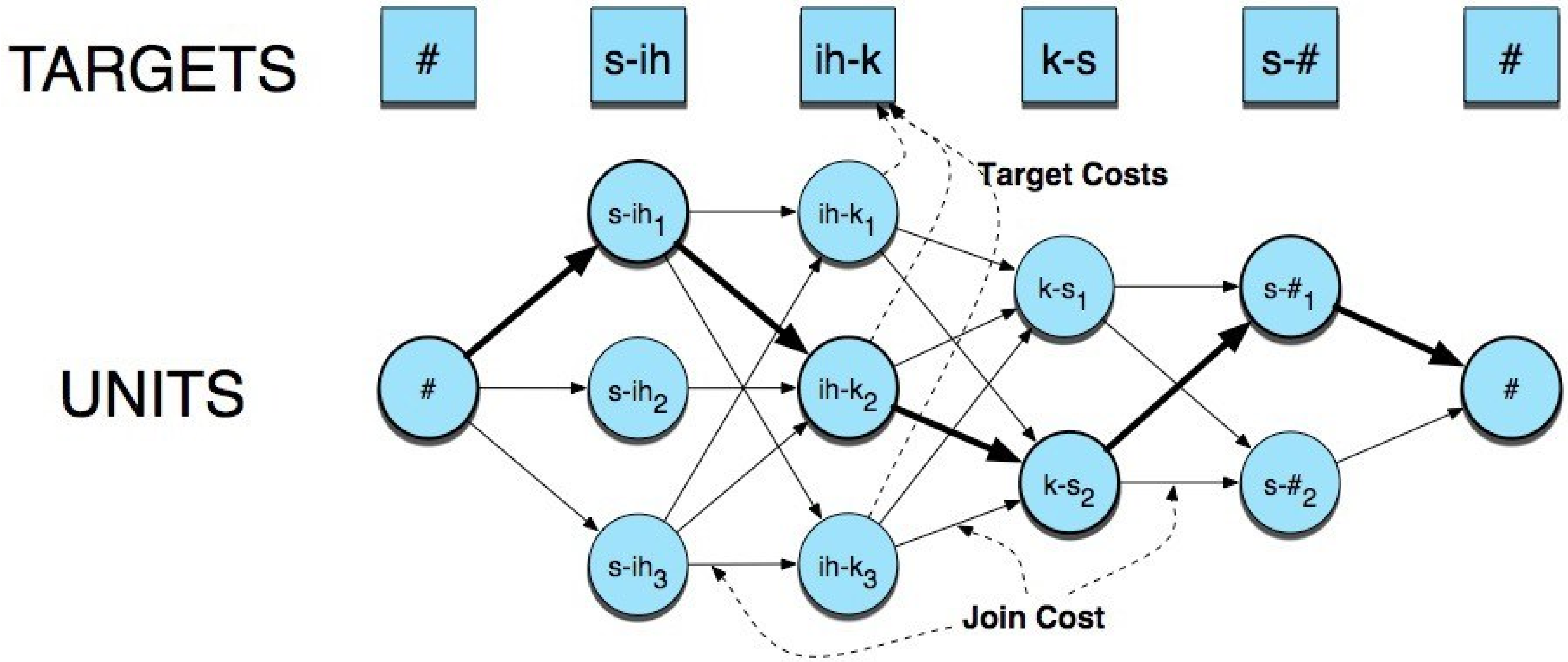
Architecture
Target & Join Costs

- Problems of diphone synthesis
  - Signal processing leaves artifacts
  - Only captures local effects (neighboring phones)
- Unit selection synthesis
  - Larger units (from diphones to sentences)
  - 10 hours of speech instead of 1172 diphones (a few minutes)
    - If words or sentences not present in database, likely to find fitting diphone due to many copies of diphone
  - Little signal processing applied to each unit

Unit Selection Synthesis

- For each diphone s_t that we want to synthesize, find the unit u_j in the database that minimizes target and join costs
- Target cost $T(s_t, u_j) = \sum_{n=1}^{N} w_n T_n(s_t, u_j)$
  - Closest match to target description
  - n features (weighted): F0, stress, duration, position,…
- Join cost $J(u_j, u_{j+1}) = \sum_{p=1}^{P} w_p J_p(u_j, u_{j+1})$
  - Best join with neighboring (next) unit,
  - p features: F0, energy, spectral features,…
  - $J(u_j, u_{j+1}) = 0$ if u_j and u_{j+1} consecutive diphones in database

Unit Selection Search



$$\hat{U} = \underset{U}{\operatorname{argmin}} \sum_{j=1}^{J} T(s_j, u_j) + \sum_{j=1}^{J-1} J(u_j, u_{j+1})$$

Summary

- Common recipe for any TTS effort to achieve best results (data + evaluation are critical)
- Concatenative systems
  - Easier to get working quickly (no modeling work)
  - Low level signal processing and joins cause artifacts
  - Require large training sets for best coverage
- Editing prosody is critical for human-like TTS
  - No natural way to do this in concatenative systems